

PRAKTIKUM SISTEM OPERASI

MODUL 4

MEMBACA SOURCE CODE XINU



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

2026

JURNAL

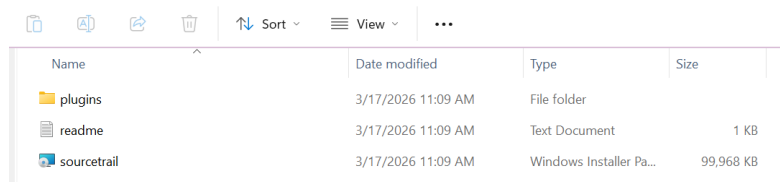
1. [10 Poin] Apa nama image yang dihasilkan setelah melakukan kompilasi pada Xinu? Berapa ukuran file tersebut? Ada pada folder apa file image tersebut?

Hint: baca kembali modul-modul sebelumnya

Jawaban: Image yang dihasilkan biasanya memiliki nama xinu.elf atau xinu.bin, tergantung pada konfigurasi yang digunakan selama proses build. File tersebut umumnya disimpan di dalam direktori xinu/compile/, dengan ukuran yang bervariasi, berkisar sekitar ratusan kilobyte (KB), sesuai dengan hasil build dan konfigurasi yang diterapkan.

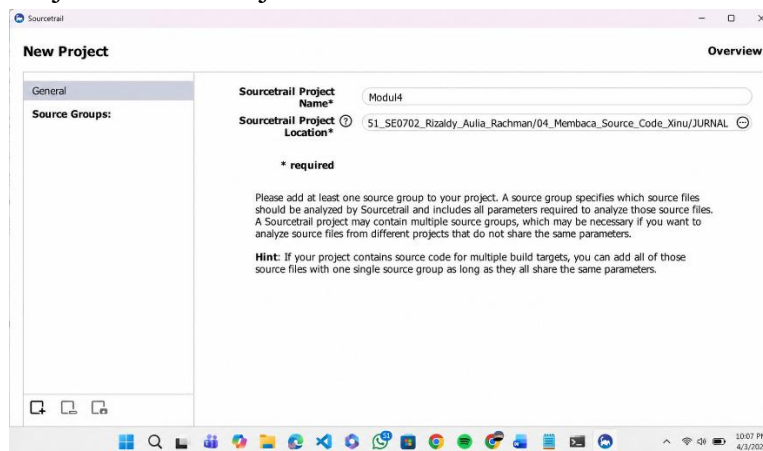
2. Membaca source code Xinu

- a. Cek aplikasi bernama Sourcetrail di PC. Jika belum ada, download SourceTrail pada (DOWNLOAD SOURCETRAIL). SourceTrail adalah software untuk mengeksplorasi source code. Programmer yang bagus lebih banyak membaca kode daripada menulis kode

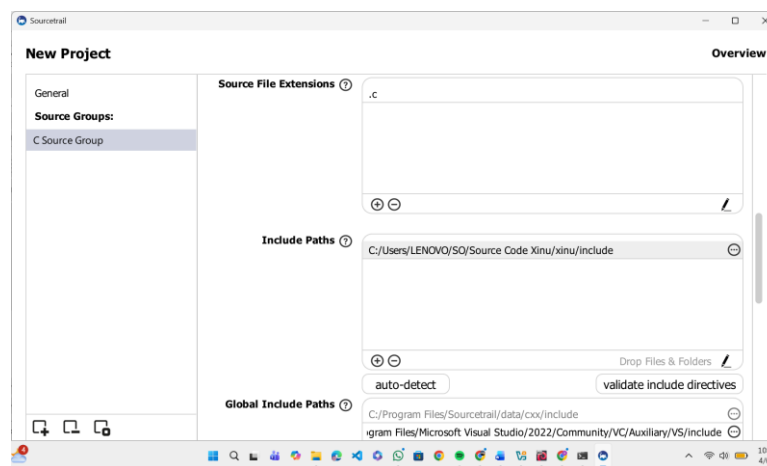
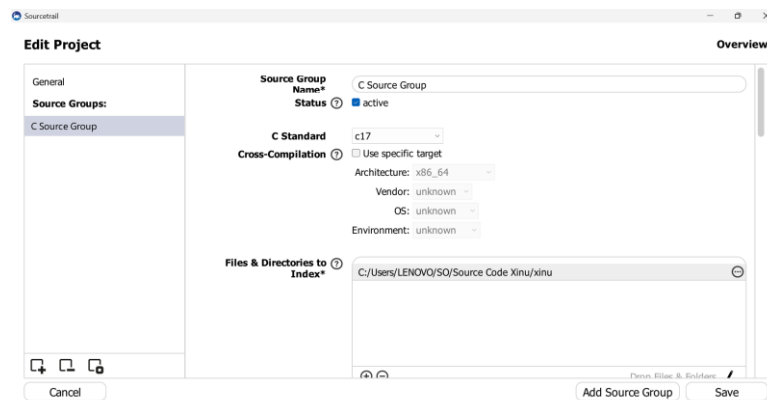


Name	Date modified	Type	Size
plugins	3/17/2026 11:09 AM	File folder	
readme	3/17/2026 11:09 AM	Text Document	1 KB
sourcetrail	3/17/2026 11:09 AM	Windows Installer Pa...	99,968 KB

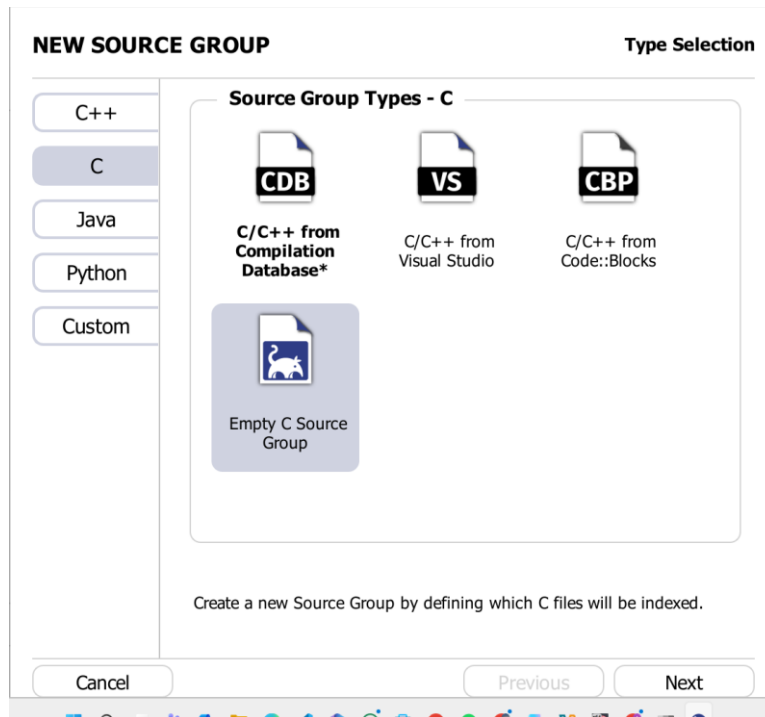
- b. Jalankan SourceTrail
- c. Project → New Project



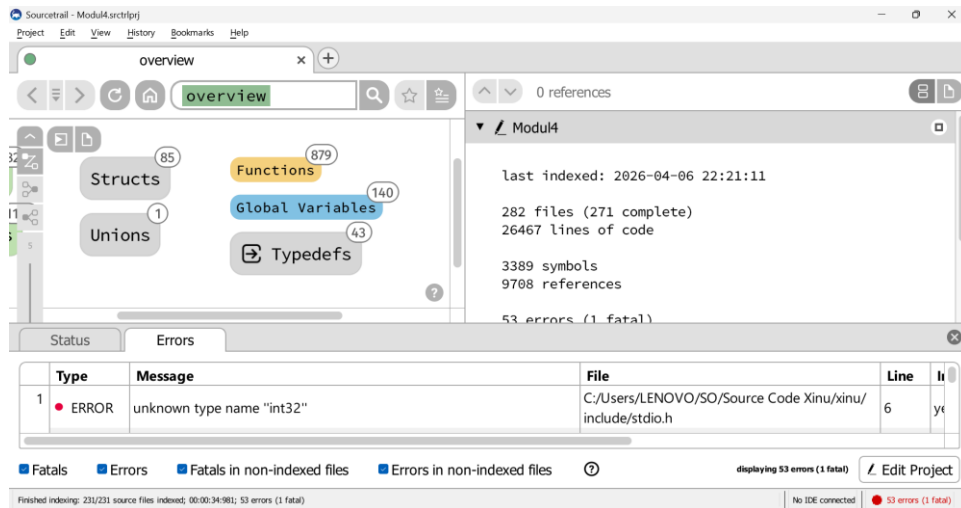
- d. Isi nama project xinu dan pilih lokasi project di manapun



- e. Add Source Groups, pilih C, lalu pilih Empty C Source Group



- f. File & Directories to Index: masukkan semua folder Xinu (yang sebelumnya telah di download)
- g. Include Paths: .../xinu/include
- h. Create
- i. Silahkan eksplorasi source code Xinu



Sourcetrail - Modul4.srctrl.prj

Project Edit View History Bookmarks Help

rdspcress.c

rdspcress.c

1 references

rdspcress.c

```

1 /* rdspcress.c - rdspcress */
2
3 #include <xinu.h>
4
5 /*-----
6  * rdspcress - High-priority background process tha
7  * extracts an item from the request queue, sen
8  * request to the remote disk server, and handl
9  * response, including caching responses blocks

```

Status Errors

Type	Message	File	Line	li
1	ERROR	unknown type name "int32"	6	ye

Fatals Errors Fatals in non-indexed files Errors in non-indexed files

displaying 53 errors (1 fatal) / Edit Project

Activate "/.../Source Code Xinu/xinu/device/rdspcress.c": 1 result with 0 references in 1 file

No IDE connected 53 errors (1 fatal)

Sourcetrail - Modul4.srctrl.prj

Project Edit View History Bookmarks Help

create.c

create.c

1 references

create.c

```

1 /* create.c - create, newpid */
2
3 #include <xinu.h>
4
5 local int newpid();
6
7 /*-----
8  * create - Create a process to start running a fun
9  *

```

Status Errors

Type	Message	File	Line	li
1	ERROR	unknown type name "int32"	6	ye

Fatals Errors Fatals in non-indexed files Errors in non-indexed files

displaying 53 errors (1 fatal) / Edit Project

Activate "/.../Source Code Xinu/xinu/system/create.c": 1 result with 0 references in 1 file

No IDE connected 53 errors (1 fatal)

Sourcetrail - Modul4.srctrl.prj

Project Edit View History Bookmarks Help

kill.c

kill.c

1 references

kill.c

```

1 /* kill.c - kill */
2
3 #include <xinu.h>
4
5 /*-----
6  * kill - Kill a process and remove it from the syst
7  *
8  */
9 syscall kill(

```

Status Errors

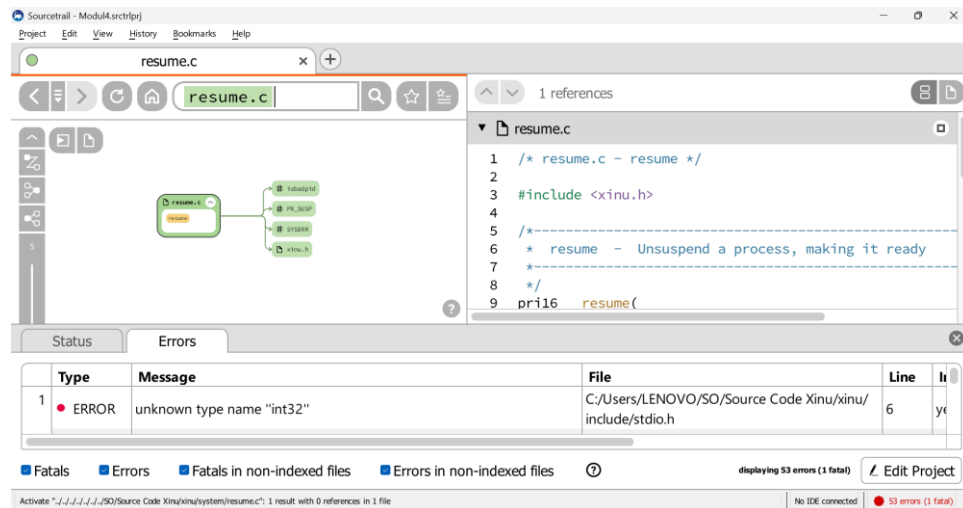
Type	Message	File	Line	li
1	ERROR	unknown type name "int32"	6	ye

Fatals Errors Fatals in non-indexed files Errors in non-indexed files

displaying 53 errors (1 fatal) / Edit Project

Activate "/.../Source Code Xinu/xinu/system/kill.c": 1 result with 0 references in 1 file

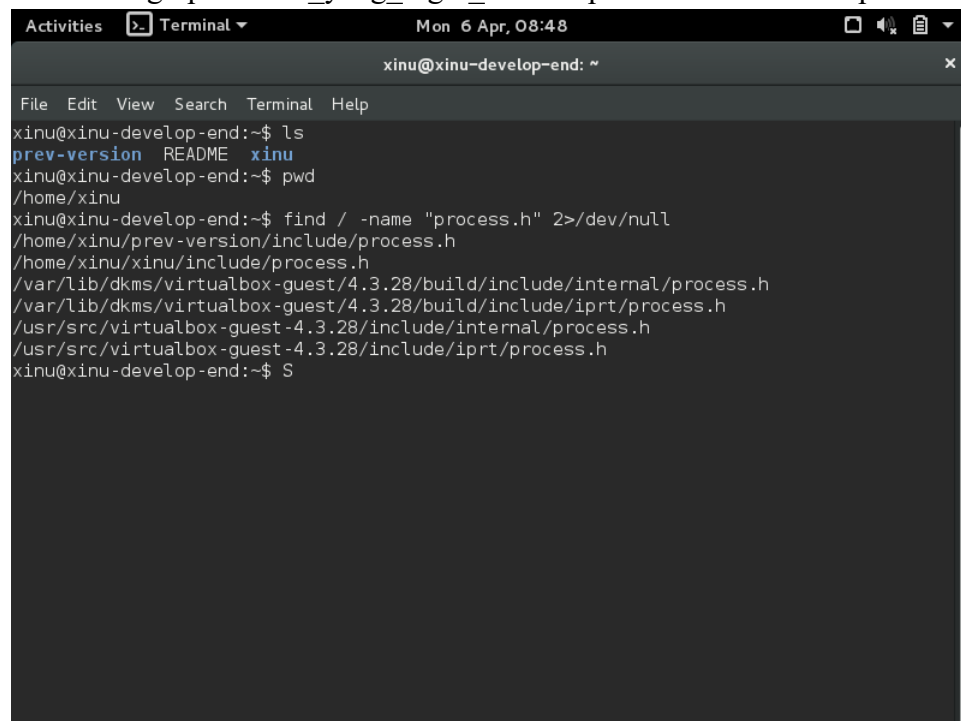
No IDE connected 53 errors (1 fatal)



3. [10 Poin] Carilah struktur data dari proses pada Xinu OS. Struktur data proses ada pada file apa? Informasi apa saja yang disimpan dalam struktur data tersebut? Hint: file berektensi .h

Pada Linux bisa digunakan perintah “grep”

Contoh: `$ grep -r “kata_yang_ingin_dicari” /path/ke/direktori/tempat/file/berada`



```
Activities Terminal Mon 6 Apr, 08:53
xinu@xinu-develop-end: ~

File Edit View Search Terminal Help
xinu@xinu-develop-end:~$ ls
prev-version  README  xinu
xinu@xinu-develop-end:~$ pwd
/home/xinu
xinu@xinu-develop-end:~$ find / -name "process.h" 2>/dev/null
/home/xinu/prev-version/include/process.h
/home/xinu/xinu/include/process.h
/var/lib/dkms/virtualbox-guest/4.3.28/build/include/internal/process.h
/var/lib/dkms/virtualbox-guest/4.3.28/build/include/iprt/process.h
/usr/src/virtualbox-guest-4.3.28/include/internal/process.h
/usr/src/virtualbox-guest-4.3.28/include/iprt/process.h
xinu@xinu-develop-end:~$ grep -r "banner" /home/xinu/xinu/include
/home/xinu/xinu/include/shell.h:/* Shell banner (assumes VT100) */
xinu@xinu-develop-end:~$ grep -r "banner" /home/xinu/xinu/shell/
/home/xinu/xinu/shell/shell.c: /* Print shell banner and startup message */
xinu@xinu-develop-end:~$ grep -r "struct proctent" /home/xinu/xinu/include/
/home/xinu/xinu/include/process.h:struct proctent { /* Entry in the process
table */
xinu@xinu-develop-end:~$ cat /home/xinu/xinu/include/process.h
/* process.h - isbadpid */

/* Maximum number of processes in the system */

#ifdef NPROC
#define NPROC 8
#endif

/* Process state constants */
```

```
Activities Terminal Mon 6 Apr, 08:53
xinu@xinu-develop-end: ~

File Edit View Search Terminal Help
/* Process state constants */

#define PR_FREE 0 /* Process table entry is unused */
#define PR_CURR 1 /* Process is currently running */
#define PR_READY 2 /* Process is on ready queue */
#define PR_RECV 3 /* Process waiting for message */
#define PR_SLEEP 4 /* Process is sleeping */
#define PR_SUSP 5 /* Process is suspended */
#define PR_WAIT 6 /* Process is on semaphore queue */
#define PR_RECTIM 7 /* Process is receiving with timeout */

/* Miscellaneous process definitions */

#define PNMLEN 16 /* Length of process "name" */
#define NULLPROC 0 /* ID of the null process */

/* Process initialization constants */

#define INITSTK 65536 /* Initial process stack size */
#define INITPRI0 20 /* Initial process priority */
#define INITRET userret /* Address to which process returns */

/* Inline code to check process ID (assumes interrupts are disabled) */

#define isbadpid(x) ((pid32)(x) < 0) || \
((pid32)(x) >= NPROC) || \
(proctab[(x)].prstate == PR_FREE))
```

4. **[80 poin]** Mengubah welcome banner pada Xinu!
 - a. Carilah file yang menyimpan banner Xinu!
Hint: file berekstensi .h pada direktori xinu/include
 - b. Carilah file yang menampilkan banner Xinu!
Hint: file berekstensi .c pada direktori xinu/shell
 - c. Ubahlah welcome banner Xinu sehingga menjadi seperti ini:
Tambahkan NAMA dan NIM masing-masing, kemudian ubah banner Xinu sesuai dengan selera Anda

- d. Setelah melakukan perubahan lakukan perintah berikut ini pada terminal Linux
- `$ cd xinu/compile`
 - `$ make clean`
 - `$ make`
 - `$ sudo minicom`
 - Jalankan `vm backend`


```
Activities Terminal Mon 13 Apr, 20:48
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help
4447448 bytes of free memory. Free list:
    [0x0015E320 to 0x0009FFF7]
    [0x00100000 to 0x005FBFFF]
115657 bytes of Xinu code.
    [0x00100000 to 0x0011C3C8]
148488 bytes of data.
    [0x0011FEC0 to 0x001442C7]

Obtained IP address 10.0.3.15 (0x0a00030f)

-----RIZALDY AULIA RACHMAN-----

XINU

-----2311104051-----

Welcome to Xinu!

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0
```